

PH 582: Assignment-2

Kane-Mele Model Numerical Simulations

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Note: Parameters are taken in arbitrary units for numerical solution, e.g. nearest neighbor distance is taken as 1, and unless otherwise stated, t is set to 1. The onsite energy shift is taken to be 0 but on-site energy offset is taken. If the question did not ask to plot for various parameters, a single set of parameters as stated in the report will have been used for the plot. Jupyter notebook was used for plotting using python and markdown cells can be used to find the code for any plot. Also the following short forms might be used at various places, both in report and the code:

- NN: Nearest neighbor (t : Hopping parameter)
- BS: Band structure
- Ω : Berry curvature
- $\mathcal{C}$: Chern number
- TBD: Topological band diagram
- IS: Inversion symmetry
- TRS: Time reversal symmetry
- SOC: Spin-Orbit Coupling
- BZ: Brillouin Zone
- SHE: Spin-Hall effect
- λ_{SOC} : Spin-orbit coupling parameter
- λ_v : Mass term
- λ_R : Rashba term parameter

The Hamiltonian for Kane-Mele model is given below. This is considered in answers, wherever required.

$$H = t \sum_{\langle ij \rangle} c_i^\dagger c_j + i\lambda_{SO} \sum_{\langle\langle ij \rangle\rangle} \nu_{ij} c_i^\dagger s_z c_j + i\lambda_R \sum_{\langle ij \rangle} c_i^\dagger (\mathbf{s} \times \hat{\mathbf{d}}_{ij})_z c_j + \lambda_v \sum_i \xi_i c_i^\dagger c_i \quad (1)$$

1. **Full BZ Tight-Binding Modell:** The band structure of tight-binding honeycomb model with Rashba term is shown in Fig.1. $\lambda_{SOC} = 0$ is assumed as it is considered in further questions.

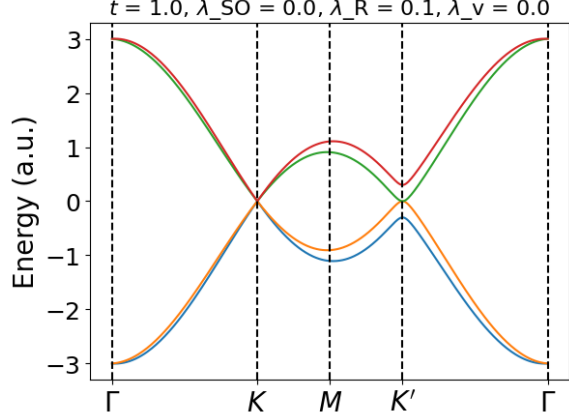


Figure 1: Rashba term is added here along with hopping. We see spin-splitting at and around K' point.

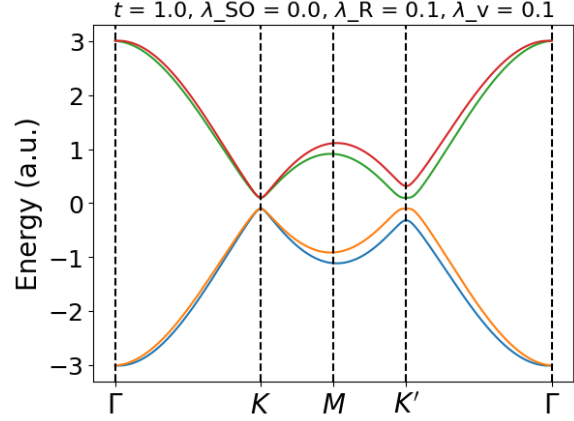


Figure 2: Here, mass term (λ_v) is made non-zero to see the opening of band-gap.

2. **Spin-Hall Effect:**